

FOURIER ANALYSIS KIT®

OBJECTIVE

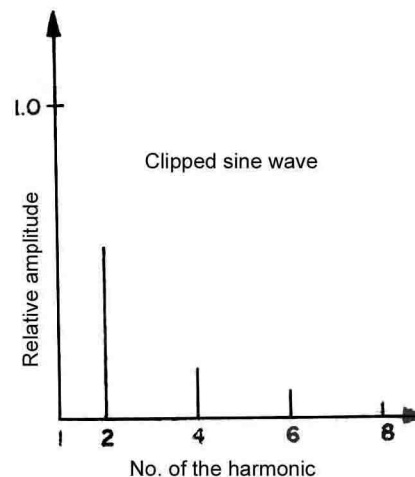
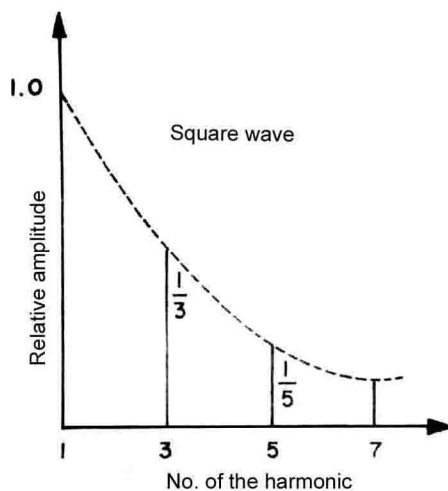
TO Analyze complex wave (square, clipped sine wave triangular wave etc.)

TO verify the existence of different harmonics and measure their relative amplitudes.



In electronics, we come across complex signals. They consist of pure sine and cosine waves. The mathematical tool that helps us in the analysis of complex signals is called the Fourier Theorem. According to this theorem, a square wave consists of the fundamental and other odd harmonics. Similarly, a clipped sine wave consists of the fundamental and other even harmonics.

Typical results obtained with the kit are shown below :



INSTRUMENT

The Kit consists of a stabilized power supply (6-0-6), a function generator and the analyzer. Square, triangular and clipped sine waveforms are generated by a specially designed integrated circuit function generator whose output frequency can be varied in the range of 500 Hz to 15 KHz using the 100KΩ Ten Turn potentiometer. The output level of the sine and triangular waves can be varied using the amplitude control provided on the front panel of the unit.

The only equipment needed for the experiment is any general purpose C.R.O. available in laboratory.

Manufacturers:



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